MVE Implementation Spec v1

Version 0 technical specification for experiment-ready cross-subject shared perceptual cell learning on NSD

Version 0 objective
Lock the first experiment to one falsifiable claim: a learned shared cell space should improve held-out-subject closed-set stimulus identification on NSD relative to strong non-graph baselines. Version 0 intentionally freezes the scope before graph reasoning.

Version boundary. This document replaces proposal-level ambiguity with coding-level defaults. It assumes NSD only, the shared 1000-image stimulus subset, averaged repeated-image responses, no dynamic graph module, and a single primary metric for the first pass.

1. Sample Definition

Item	Version 0 decision
Dataset	NSD only for primary development and acceptance testing.
Sample unit	One subject x one shared image, using the subject's averaged repeated-image beta response for that image.
Label	Closed-set image identity over the 1000 NSD shared stimuli. This keeps the first metric simple and directly tied to cross-subject alignment.
Split	Outer loop = 8 leave-one-subject-out folds. Per fold: 1 held-out test subject, 1 validation subject, 6 training subjects.
Train/val/test policy	No stimulus split in Version 0. All 1000 shared images appear in every split; only the subject axis changes. This isolates subject transfer rather than open-set semantic generalization.

Rationale. Averaging repeated presentations reduces trial noise and lets the first implementation answer the alignment question before single-trial variance, temporal modeling, or external datasets are introduced.

2. Input Representation

Component	Specification
Signal level	Voxel-level beta pattern restricted to selected visual ROIs; no ROI mean pooling for the main model.
ROIs	NSD visual ROI union: V1, V2, V3, hV4, LO, VO, and PHC masks from official NSD ROI annotations.
Raw tensor	Subject-specific vector with all voxels in the selected ROI union before projection.
Normalization	Fit voxelwise z-score statistics on training subjects only; apply the fitted mean/std to val and test subjects per fold.
Projection	Subject-wise PCA fitted on training samples only, projected to d_in = 512. No whitening in Version 0.
Model input shape	[512] per sample after projection; batch shape [B, 512].

Stored artifacts. Keep both the raw ROI voxel indices and the fitted PCA objects so that later diagnostics can map prototypes back to anatomy without rerunning extraction.

3. Version 0 Model Definition

Architecture. The proposed model is static and sequence-free in Version 0: subject encoder -> shared prototype bank -> cell activation vector -> classifier head. Dynamic GNN blocks are explicitly deferred.

Module	Default setting
Subject encoder f_s	Subject-specific two-layer MLP: 512 -> 256 -> 256 with GELU, dropout 0.1, and LayerNorm on the output.
Shared prototype bank P	P in R^(K x d), with K = 64 shared cells and d = 256 embedding dimensions.
Assignment rule	Cosine similarity followed by softmax with temperature tau = 0.07. No Sinkhorn and no hard top-k masking in Version 0.
Cell activation z	K-dimensional activation vector z = a. The model uses one activation vector per sample, not a T x K sequence.
Output head g	LayerNorm -> Linear(64, 256) -> GELU -> Linear(256, 1000).

```
x_s in R^512
h = f_s(x_s) in R^256
u_k = cos(h, p_k)
a = softmax(u / tau),  tau = 0.07
z = a in R^64
y_hat = g(z)
L = L_ce + 0.20 L_align + 0.05 L_balance + 0.01 L_orth
```

L_align. For each image represented by multiple training subjects in a batch, minimize 1 - cosine(z_i, z_j) across subject pairs. **L_balance.** Match the batch-mean assignment distribution to Uniform(64) with KL divergence. **L_orth.** Penalize prototype redundancy with $||PP^T - I||^2_F$.

4. Baseline Hierarchy

ID	Baseline	Purpose
B1	ROI feature + linear classifier	Lowest-friction atlas-aware baseline on the same 512-d projected input.
B2	ROI feature + two-layer MLP	Nonlinear baseline without shared cell structure.
B3	Shared linear latent space	Subject-specific linear adapters to 256-d shared latent plus linear classifier.
B4	Shared encoder without prototype bank	Strong non-graph baseline: same subject encoder as the proposed model, but h goes directly to the classifier head.
B5	Prototype bank model	Proposed Version 0 model and the only new method required for acceptance.

Decision rule. Graph modules will not be introduced unless B5 beats both B3 and B4 on held-out subjects. Beating only weak ROI baselines is not sufficient.

5. Training Protocol

Item	Version 0 default
Batch construction	Sample 32 image IDs per step and draw up to 4 training subjects per image; target batch size is 128.
Subject mixing	Yes. Every batch mixes subjects so that L_align is defined within the batch.
Optimizer	AdamW with parameter groups.
Learning rates	1e-3 for subject encoders and classifier head; 5e-4 for prototype bank.
Regularization	Weight decay 1e-4, dropout 0.1, gradient clipping at global norm 1.0.
Schedule	5 warmup epochs then cosine decay to 1e-5.
Epochs	80 max epochs.
Early stopping	Patience 10 on validation top-1 accuracy; break ties with validation top-5 accuracy.
Seeds	5 fixed seeds: 11, 22, 33, 44, 55.

6. Evaluation Protocol

Level	Metric or report
Primary	Held-out-subject top-1 closed-set accuracy over the 1000 shared images.
Secondary	Top-5 accuracy, retrieval Recall@1 and Recall@5 from the penultimate embedding, and RSA between subject-level representation dissimilarity matrices.
Consistency analysis	Cell consistency = mean same-image cross-subject cosine on z plus top-1 prototype agreement rate.
Reporting	Mean +/- std across 8 subject folds and 5 seeds, with per-subject tables retained in the results directory.
Acceptance gate	Prototype model should exceed B4 by at least 3 absolute points on mean top-1 accuracy or win on at least 6 of 8 held-out subjects.

External validation. THINGS-fMRI and BOLD5000 remain frozen-transfer checks after NSD succeeds. They are not part of the Version 0 acceptance gate, but the evaluation scripts should already support loading an external feature matrix and running a linear probe on frozen z.

7. Failure Diagnostics

Failure mode	Detection signal	First response
Prototype collapse	exp(H(mean(a))) < 16 or one cell gets > 25% of assignments.	Raise L_balance, reduce K to 32, inspect assignment histograms.
Cross-subject misalignment	Same-image cosine on z is not at least 0.10 above different-image cosine.	Audit z-scoring, PCA fitting, and subject-mixed batches first.
Overfitting	Train top-1 is > 10 points above validation for 5 straight epochs.	Increase dropout, shrink hidden width, and stop adding objectives.
Weak external transfer	NSD wins but frozen transfer to THINGS/BOLD5000 is unstable.	Check ROI overlap, preprocessing leakage, and label mapping before claiming dataset shift.

8. First Coding Milestones

Module	Output artifact
Data manifest builder	data/nsd_v0/shared1000_manifest.csv and subject-fold split JSON files.
ROI extraction and PCA	data/nsd_v0/features/subjXX_visual_pca512.npz and artifacts/nsd_v0/preprocess/subjXX_pca.pkl.
Baseline training script	results/nsd_v0/baselines/{model}/fold_{k}_seed_{s}/metrics.json and checkpoint.pt.
Prototype model trainer	results/nsd_v0/prototype/fold_{k}_seed_{s}/model.pt, metrics.json, assignment_stats.json.
Evaluation script	results/nsd_v0/eval/summary.csv and subject_breakdown.csv.
Visualization script	results/nsd_v0/figures/prototype_usage.png, alignment_heatmap.png, subject_rsa.png.

Implementation order

Week 1 should end with the NSD shared-stimulus manifest and projected features on disk. Week 2 should end with B1-B4 training runs and a stable evaluation script. The prototype model should start only after those artifacts exist.

Repository-ready checklist. Every milestone above should map to one config file, one executable script, and one machine-readable output. If a milestone only produces narrative notes, it is not complete.

Deferred items. Single-trial beta modeling, temporal bins, graph edges between cells, and external-dataset fine-tuning are explicitly outside Version 0. If any of these appear in the first implementation PR, the scope has drifted.

This specification is experiment-ready when every table entry above can be mapped to a config key, a saved artifact, or a unit-tested dataloader assumption.